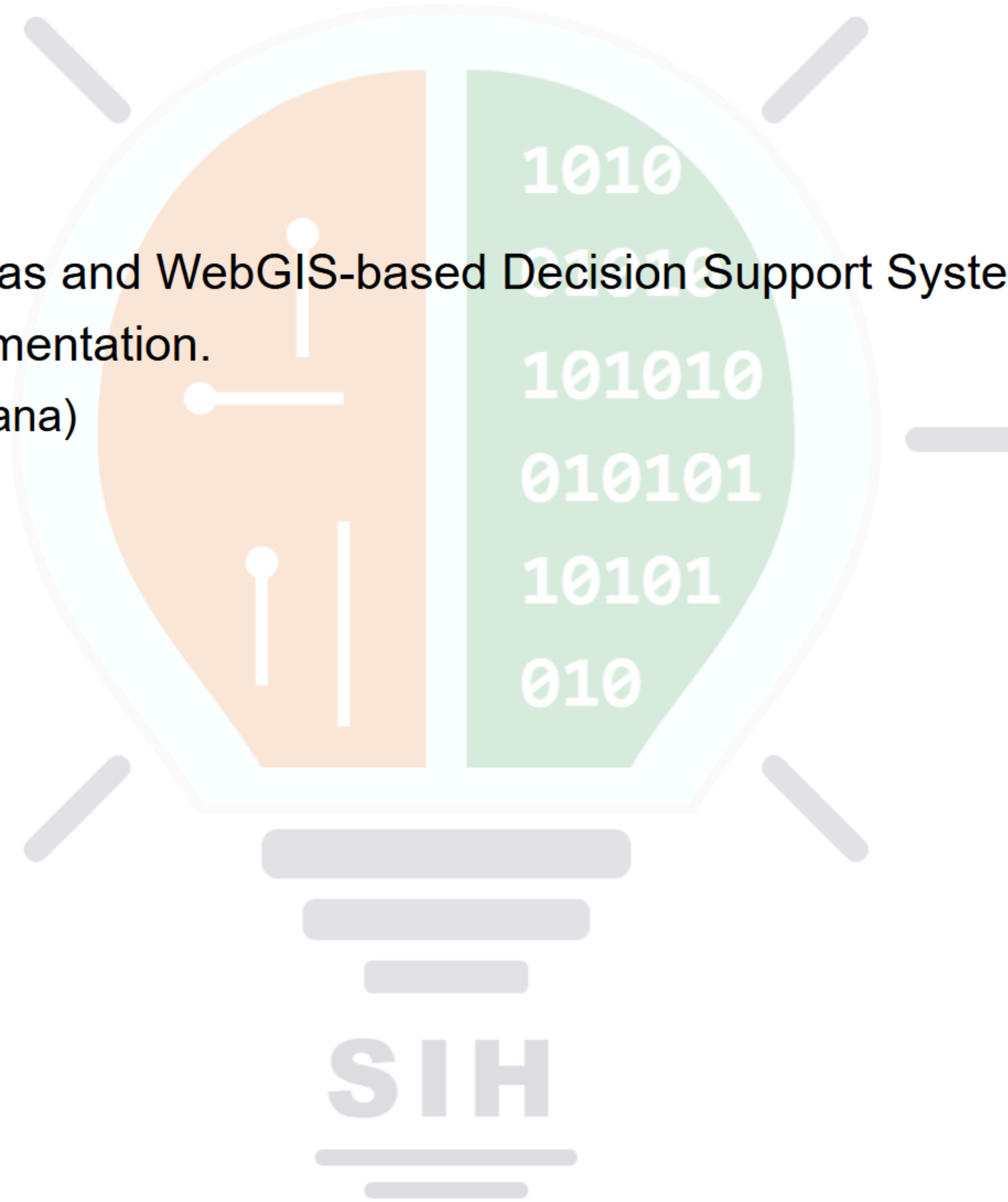


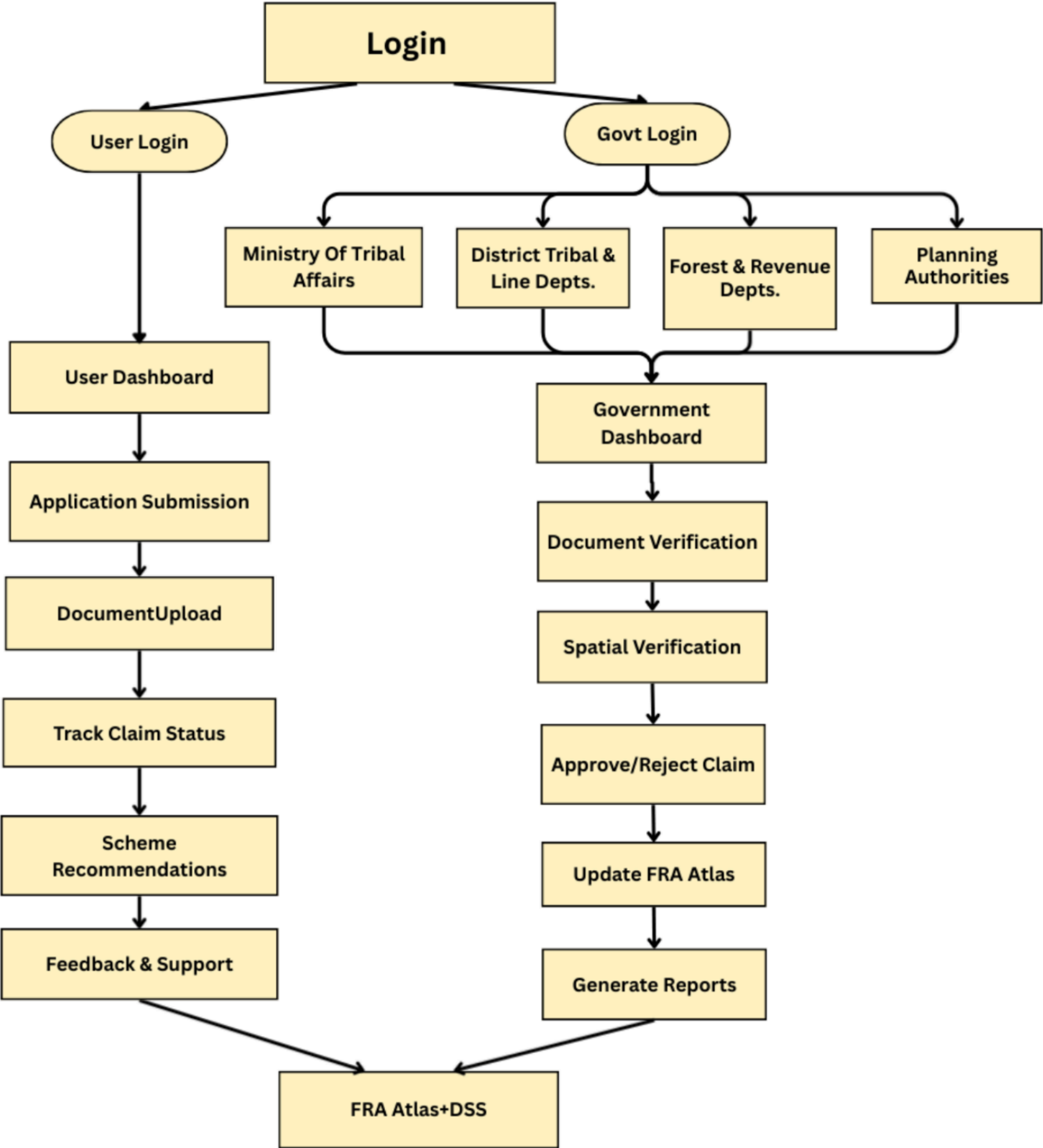
TITLE PAGE

- **Problem Statement ID** –SIH25108
- **Problem Statement Title**-Development of AI-powered FRA Atlas and WebGIS-based Decision Support System (DSS) for Integrated Monitoring of Forest Rights Act (FRA) Implementation.
(States to be concentrated: Madhya Pradesh, Tripura , Odisha, Telangana)
- **Theme**-Miscellaneous
- **PS Category**- Software
- **Team ID**- 57385
- **Team Name**- Team Hackastra

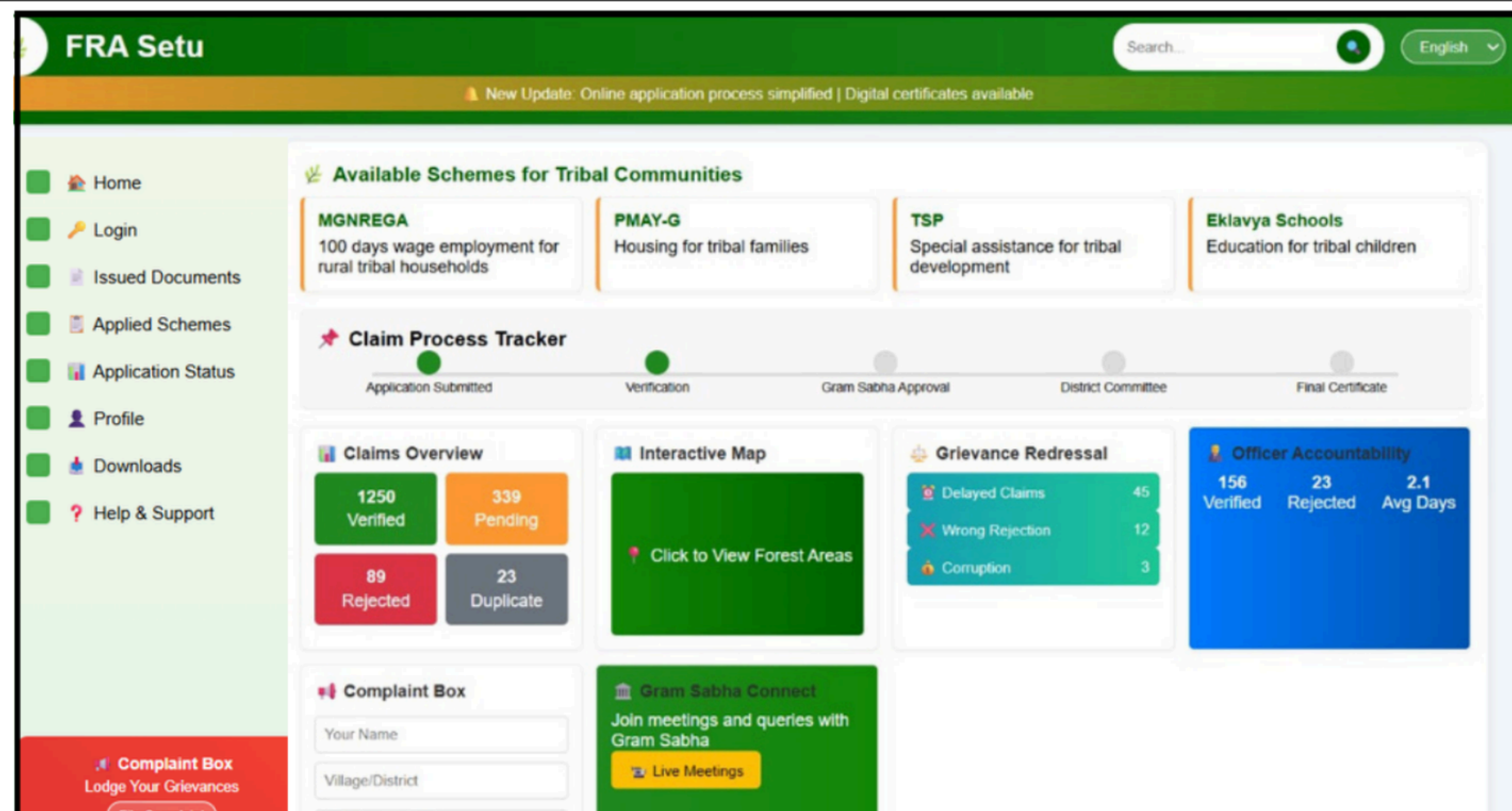


Proposed Solution:

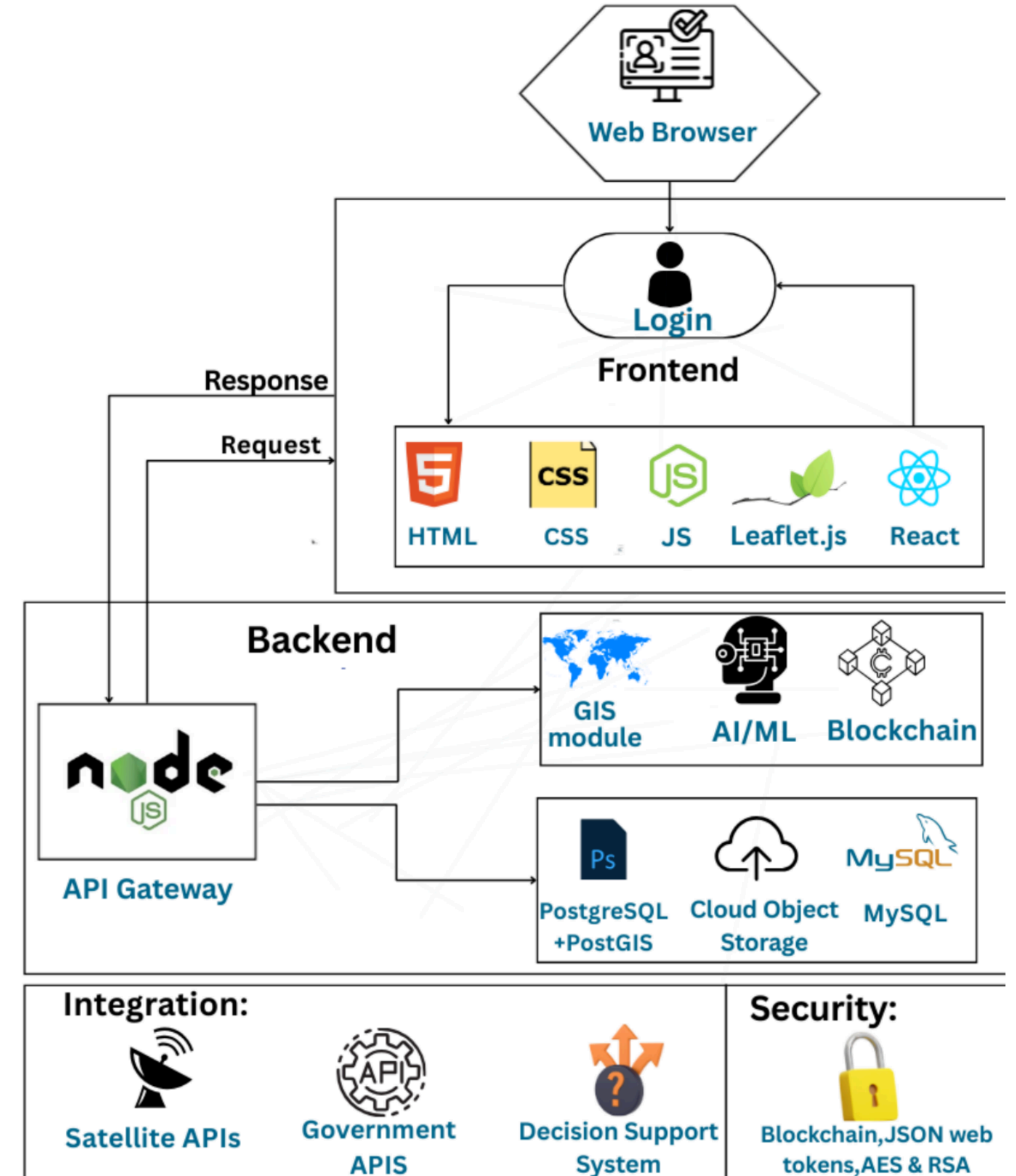
- NGO Digital Aid :**
Teach Gram Sabhas and communities apps, IVR, and GIS tools.
- Digital Witness & Testimony:**
Voice and video records stored securely as valid proof.
- Title Insurance / Digital Patta :**
Ensures authenticity with secure 7/12 land record entry.
- Grievance Resolution & Illegal Alerts:**
Efficient FRA grievance handling and unauthorized land encroachment warnings.
- AI Disciplinary Dashboard :**
Suggests actions and tracks officer accountability.
- Analytics-Driven Officer Monitoring:**
Tracks verification time, claim outcomes, and grievances reported.
- Interactive FRA Web-GIS Atlas:**
Dynamic platform for exploring forest rights and assets.
- AI Disputes Mediator:**
Resolving conflicts within existing legal frameworks.
- GI Tagging and E-Signature :**
To preserve cultural heritage safeguard traditional products and paperless approval .



- **Data Extraction:** Optical Character Recognition(OCR), NLP.
- **WebGIS platform:** Leaflet.js(for interactive maps).
- **Asset Mapping:** Random Forest & CNN.
- **Data Storage:** PostgreSQL & PostGIS(for geospatial data).
- **Data Management:** Cloud Object Storage & MySQL(Structured).
- **GeoServer:** Use standard protocols WMS and WFS to serve maps.
- **Decision Support System:** for scheme recommendations.
- **Blockchain based Security:** Hyperledger Fabric.
- **Key Lock Framework:** Open-source Identity & Access Management
- **Rasa:** Preferred for its offline, multilingual, and secure chatbot.



User Interface: It is a draft version, subject to future updates.



❖ Feasibility:

1. Technical Feasibility:

1. Digitization, AI Mapping and Web-GIS Dashboards:

- OCR and AI extract data from documents.
- Web-GIS dashboards show real-time interactive maps.

2. DSS Integration with Blockchain Security:

- AI engines match schemes to community needs.
- Blockchain ensures secure, tamper-proof digital records.

2. Operational Feasibility:

1. Existing Setup and Simple Staff Training:

- Government data exists, needs proper connection.
- Staff need basic training through short workshops.

2. Stakeholder Support and Effective FRA Implementation:

- Ministries and NGOs already support FRA schemes.
- System strengthens efforts, not replacing existing work.

3. Legal & Ethical Feasibility:

1. Data Privacy:

- Must follow India's DPDP Act, ensuring legal compliance.
- Blockchain and encryption safeguard sensitive user data.

2. Consent Mechanism:

- FRA holders' informed consent required in all processes.
- Real-time feedback enables secure two-way communication.

❖ Viability:

1. Economic Viability:

1. Cost-Effective & Efficient:

- Saves paperwork, effort, duplication, and reduces fraud.
- Cuts processing delays, expenses, and operational overhead.

2. Supported & Scalable Technology:

- Backed by government programs and tribal inclusion schemes.
- Open-source tech ensures flexible, low-cost, future-ready deployment.

2. Social Viability:

1. Empowers Communities & Improves Scheme Access:

- Provides transparency, secure records, better documentation.
- Directly benefits FRA holders through scheme access.

2. Reduces Conflict & Promotes Stronger Inclusion:

- Verified records, alerts reduce frequent land disputes.
- Mobile updates, two-way communication improve engagement.

3. Scalability:

1. Pan-India Expansion for Wider Community Benefits:

- Pilot in districts, then scale across India.
- Adjust platform easily for local data.

2. Cross-Sector Use of WebGIS + DSS:

- Adaptable for agriculture, urban development needs.
- Useful in disaster management and planning.

❖ Benefits:

1.Social Benefits:

- Digital platform enables communities with land rights tracking.
- IVR helpline, e-learning ensure inclusive literacy access.
- Transparent records, video testimonies reduce conflict, build trust.

2. Environmental Benefits:

- Sustainable forest use by recognized communities.
- Conservation through community stewardship.
- GI Tagging preserves cultural heritage & traditional products.

3. Economic Benefits:

- Access to schemes (MGNREGA, PM-KISAN, DAGAJU, Jal Jeevan Mission)
- GI Tagging boosts local products & tribal livelihoods
- Title insurance & digital patta ensure secure land ownership → easier loans & income stability

4. Governance Benefits:

- Audit trails ensure accountability, prevent misuse.
- Whistleblowing portal boosts transparency, fights corruption.
- Cross-validation stops duplication and fraudulent claims.

5.Legal & Security Benefits:

- E-Signature & Digital Patta provide legally valid ownership records.
- Protection against land encroachment (Illegal Practice Alert).
- Digital witness/testimony ensures fair dispute settlement.

❖ Impacts:

1.Secure Land Rights & Transparency:

- Communities track claims, reducing corruption and fraud.
- Digital patta ensures authenticity in land ownership records.
- Builds trust between citizens and government authorities.

2. Forest Protection & Conservation with AI:

- AI monitors land to prevent illegal encroachment.
- Conserves biodiversity and safeguards natural resources.
- Promotes sustainable, community-driven forest management.

3. Boost Livelihoods & Local Development:

- Resource planning improves incomes and rural economy.
- GI tagging protects heritage, boosts local products.
- FRA enables fair access to welfare schemes.

4. Digital, Eco-Friendly & Cost Effective:

- Reduces paperwork,ensuring greener, eco-friendly governance.
- Preserves tribal knowledge through digital archiving tools.
- Lowers administrative costs and increases efficiency.

5. Smart, Transparent & Scalable Governance:

- Transparent dashboards track officer performance and claims.
- AI dispute mediation ensures fairness, reduces human bias.
- Unified digital platform enables future-ready expansion.

1. FRA: The Forest Rights Act (FRA), 2006 gives rights to forest-dwelling and tribal people, but challenges in implementation remain.

<https://tribal.nic.in/FRA.aspx>
<https://mpvanmitra.mkcl.org/hi>

2. Best Practices for Web Security: This guide outlines key web security measures to safeguard user information using Blockchain Technology.

<https://owasp.org/www-project-blockchain-appsec-standard/>

3. The Wildlife Protection Act: 1972 aims to protect wild animals, birds, plants, and their habitats.

<https://share.google/F3sPygTbDmIsIAmAd>

To enable smooth implementation and integration of the FRA Atlas System, the following contacts serve as official state-level resources for communication, data coordination, and support:

4. Madhya Pradesh: dirtadp@mp.gov.in

5. Odisha: <https://stsc.odisha.gov.in/acts-policies-guidelines/scheduled-tribes-and-other-traditional-forest-dwellers>

6. Tripura: <https://forest.tripura.gov.in/>

7. Telangana: <https://forestrights.telangana.gov.in/>